ActInf Textbook Group ~ Cohort 1 ~ Meeting 21 (Chapter 9, part 1)

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Hey everyone, it is October 21st, 2022.

We're in Cohort 1 of the textbook group.

It's Meeting 21, and we're having our first discussion on Chapter 9.

Chapter 9 is Model-Based Data Analysis.

Let's look at what the section headers are.

There's a short introduction, a discussion of meta-Bayesian methods, which is going to be very interesting,

and in some ways is even like an entry point to thinking about entity modeling in active inference.

And so that'll be kind of fun.

We'll return to Variational Laplace, resonating with our just-completed discussion on Chapter 4 and Laplace.

Then, Section 9.4 and 9.5 are going to help us see where data, in terms of like gigabytes of actual,

good day, Jakob, actual data from measurements and so on, where these come into play with models.

So how does one go from a furnished, trainable model to a parameterized, specified model,

which actually is explaining variance in real-world datasets.

And then there's some examples of GMs and some models of false inference.

Well, I added only one general question from a lighter reading.

I think as we all here move through this, we can generate a lot of other key points and questions.

Ultimately, the models described in this book are only useful if they can answer scientific questions.

Let's just, we could rehearse the introduction again, but let's just go into it.

So, Meta-Bayesian Methods.

This chapter deals with the utility of active inference formulations in analyzing data from behavioral experiments.

So, one could imagine all kinds of bodily, verbal, digital behavior.

This goes beyond proof of principle simulations we've seen in previous chapters and instead,

exploits active inference in answering scientific questions.

Broadly speaking, there's two related reasons for fitting a computational model to observe behavior.

The first is to estimate parameters of interest.

The second is to compare alternative hypotheses.

So, to parameterize from data is one opportunity.

That is to make some model of brain function and then understand within one person or one group or two groups

how you can consider those parameterizations to be phenotypes.

Phenotype is something of a biological organism or system that's measurable.

So, like, femur length is a phenotype.

But also, a phenotype doesn't have to just be something that's measurable on the body with a ruler.

Like, phenotype might be distance ran in the first 50 seconds after the hawk flies through the sky on a cloudy day.

And so then, that is still a measurement that could be inferred or discussed or made.

And we're talking about computational phenotyping because the data and observations for sure are like a basal phenotype.

Like, the button being clicked was measurable.

Pheno means to show.

But also, we could talk about, like, the precision variables in our cognitive model as being phenotypic.

And also, model comparison can be used.

So, whereas this first modality of parameterization is like, the dataset is fixed.

We collected 240 fMRI datasets.

Now, we're going to parameterize.

So, we're going to give some plasticity to our model.

And we're going to fit parameters so that our model is, like, the best resembling it can to these 240 fMRI datasets we have.

In the second modality, we're treating, in some ways, the models as fixed.

And then, evaluating to what extent different models stack up.

And that can be used at a very fine scale to look for different models that fit better to a given dataset.

But also, this is where we can talk about, like, actual biological explanations and predictions.

So, like, to give an example from SP...

Oh, yes, please, Ali, first.

Sorry, a related question to this whole discussion is...

I think it was in section 9.2.

It says, this goes beyond the proof-of-principle simulations we have seen in previous chapters,

and instead exploits active inference in answering scientific questions.

I didn't quite understand this statement here, because in the previous chapters,

we were also engaged with modeling the real scientific phenomena, right?

So, what does proof-of-principle simulation here exactly mean,

and how does it differ from the approach...

I mean, from answering the real scientific questions?

Yeah, good question.

Does anyone want to give a thought, or I can give a thought, too?

I mean, literally, just these...

Following that, the first and second reasons to use active inference in this way,

is, like, instead of just modeling something and saying,

oh, that looks like what we intended to model,

or, oh, gosh, it looks really predictive, or whatever,

getting more rigorous with it to specifically, you know,

enumerate parameters for a particular model, right?

And then compare multiple models, like, trying to say which one is, you know,
more precise.
That's a different thing than what was being done in the book previously.
Like, we're just building up the model and seeing how it kind of worked before.
And now we're, like, talking about putting it into, like, a machinery
that's going to, you know, select on models, like, which one's best fit or whatever.
Well said.
Yeah, I totally agree.
Preview...
Although, arguably, like, the Chapter 5, Neurobiology,
the tables of examples were answering scientific questions,
but those generative models were not largely presented.
But in the papers, one could have gone in.
But yes, the previous generative models were, like, built from the ground up.
And then, like, oh, look at this interesting behavior.
Now we're building the machinery to take in real data and fit that.
And then there's the two uses of parameterization,
or the two uses of data fitting.
To find the best parameters for a given data set and model.
And also to compare models.
Model comparison.
All right.
Now we get to the very excellent figure 9.1.
So.
Oh.
9.1 is the meta-Bayesian inference.
What does anyone see in this figure?
Ali.
Ali.
Ali.

Well, as pointed out in the text,
the inner box somehow represents the generative model of the phenomena we're trying to...
we're observing.
But the outer dashed box is somehow our generative model as the observers.
So the term meta-Bayesian, I think, used here to denote this kind of change of perspective to a higher...
to a higher level of viewpoint.

I mean, to add another layer of hierarchy to the already hierarchical generative models we had before.

Yeah.

Yeah.

Great.

Well said.

And this outer...

So we could have had an arbitrarily nested...

Well, just to make a simple point,

this is the discrete time formulation,

POMDP,

figure 4.3.

And that can be nested in an arbitrary way and composed.

And meta-Bayesian is like putting the wrapping paper on the gift.

Because this outermost layer is us.

Brock?

So the last time this came up was in the context...

I think it was in the other cohort,

but we were talking about...

the provenance of the model,

the impetus for it being...

as a tool for...

diagnosis of patients,

for non-typical sort of neurons.

So...

I guess I'm curious,

maybe,

if there's a simple way to understand it from that perspective.

But I also have...

a...

another question, I guess,

which is...

if this is...

us modeling...

It seems like it would be, like,

infinite, really recursive,

because every time you add another layer,

like, then there's another layer that's not being modeled

of how we're actually modeling it.

And therefore, we would have to...
But then if we...
accounted for that,
then...
then there would be another outer...
you know,
I don't know if that's just...
infinite regression,
towards the hidden state,
one or the other,
or...
but...
or I'm just...
that's nonsense, but...
Yeah.
No, no, it's not.
I think...
let's...
let's...
talk about the...
setting of...
the clinical neuroimaging.
So...
the inner model...
is going to be...
the cognitive behavioral model...
of the patient.
Just speaking of this...
the participant...
the recipient of action...
as the patient.
They're selecting...
green or blue coins...
or whatever.
They're doing some behavioral paradigm...
isolating...
or choice...
or decision-making.

We're modeling...

attention...

attention...

and so on.

Whether we...

formalize it or not...

the experimenter's behavior...

is influencing...

the parameterization...

of the patients.

Like...

we are choosing populations...

and sampling patients.

Here is like...

the parameterization...

of the experiment.

We'll have 20 people...

they'll come in for two sessions.

We're going to do...

11...

you know...

of this.

And here's like...

the data coming out of...

So this O here...

is the observation...

that the patient sees...

and their inference...

on what they're seeing.

Experimental stimuli...

are actually an action...

from the experimenter.

We're pushing the O...

to their screen.

So this O tilde...

is their O sequence.
We're getting...
data...
data...
from the experiment.
We could choose to...
be...
ignorant...
or implicit...
about our...
parameterization.
Or...
maybe even this...
is a nested model.
We made a...
governance decision...
in the lab...
to decide...
to decide...
who gets to decide...
this...
and the constraints...
were the cost...
and the availability.

the availability...
nested...

of the outer loop...
And...

and what about...
infinite recursion?
And I think...
one...
silver lining...
or saving grace...
or whatever...

or just...

strength of the...

or active inference...

framework...

is...

we know...

we know...

we can always...

dive in...

and...

like...

build from a node...

or we can just...

treat...

the edges...

of our base graph...

as like...

a Markov blanket...

with the unknown.

unknown.

So we could say...

we made a simple...

model of decision making...

And then we could...

this could be...

a multi-agent simulation...

and each of those...

could have another...

nested model.

And then we could...

Like if it fits...

then maybe it's irrelevant...

or...

maybe we want to know...

anyways...

but...

what...

good will it...

I don't know...

Yeah...

or it doesn't fit...

or...

Yeah...

let's just say that...

the...

the pragmatic value for us...

is...

um...

high statistical power...

and sensitivity...

for diagnosis...

of a...

neuro...

condition.

So like...

we want the inner loop...

to be...

in the point...

we...

we don't want to be like...

spending money beyond...

diminishing returns...

on our experiment...

we want to actually...

like diagnose and help...

people...

limited situation...

um...

then one could do...

model comparison...

with...

I'm ignoring all of this...
I'm just gonna...
parameterize...
just...

I'm gonna call it...
like I see it...

one could have another...
but...
but if one were to...
engage in a research program...
where we...
we collected one...
data set...
then...
yeah...
let's just call it...
like we see it...
the opportunity...
to design a second...
cohort of...
fMRI experiments...
should we do...
two people...
for a hundred sessions...
or a hundred people...
for two sessions...
who should those...
hundred people be?
so...
at that point...
having...
a...
policy selection...
outer loop...
enabling...
seems...

pretty relevant...
now...
we could make that...
simple...
maybe it's...
we always pick...
30 participants...
maybe it's...
we either do...
a small experiment...

or we do a big one...
with 30...
it's just simpler...
those are the two...
pathways we have...
or maybe it's a...
you know...
etc...
but then...
you could do...
model comparison...
and evaluation...
on those...
increasingly complex...
outer loop models...
none of that...
would be...
influencing...
the structure...
of the cognitive...
model...
of the patient...
so...
for all you...
blockference...
heads...
in the chat...

one could imagine...
that...
there's...
human...
fMRI...
cognitive model...
this is a...
versionable...
cognitive model...
open source...
and then...
different lab groups...
could have...
open or closed...
models...
with open or closed...
data...
around their...
decision making...
about how...
they're using...
a core model...
model...
this is also...
just a...
on a...
more...
qualitative point...
and then...
like...
this is where...
we can start...
to talk about...
moving beyond...
implicit biases...
not just like...
parching...
implicit biases...

or anything like that...
it's just like...
we're doing...
investigation...
and exploration...
we're externalizing...
and exteriorizing...
our priors...
and...
we're working...
such that...
the relevant features...
that are in the box...
are all...
externalized...
in terms of...
the scientific...
apparatuses...
parameterization...
and...
that's at least...
something where...

move towards...
and someone can say...
well...
you haven't made your...
hyper hyper hyper...
priors...
explicit...

isn't that a better...
conversation to have...
than...
say...

this many people...
for the experiment...

it could be like...
you didn't...
state...
the meta governance...
of how you came...
to the decision...
to do 30 participants...
I mean...
it makes it...
it sounds like...
it makes it...
really...
a lot easier...
to...
think about...
comparing...
and accounting...
showing that you...
accounted for...
you're doing...

experiment...
right...
um...
might...
be able to move...
the needle...
a little bit...
on some of the...
reproducible...
earth...
replication crisis...
things like...
in that vein...
yeah...
great...
point...
um...

um...
I think...
um...
what...
uh...
what...
uh...

mentioned...

as a side note...
uh...
I think...
approaching...
uh...

kind of...
I mean...

to...
uh...
model-based...
or...
better...
the model-based...
data analysis...
uh...
uh...
takes...
the...
infinitism...
stance...
uh...
as an implicit...
uh...
uh...
somehow...
the philosophical...

stance...
uh...
uh...
somehow...
the philosophical...

because...

if we...
uh...

in terms of...
infinitism...
I don't think...
we were able...
to...
uh...
model...
adequately...
these kinds of...
metabasean reasoning...
uh...
because...
we'll get...
at some point...
to...
uh...
as Brock...
said...
just...
uh...
infinite regressive reason...
so...
uh...
at least in my opinion...
it's somehow...
implicitly...
uh...

takes the infinitism...

uh...

stance here...

as...

uh...

proposed by Peter Klein...

uh...

at least...

uh...

uh...

uh...

uh...

uh...

uh...

will need...

somehow...

uh...

uh...

the...

subjectively...

and objectively...

available...

reasons...

uh...

uh...

to...

model...

or...

to construct...

our...

justified...

true beliefs...

uh...

in other words...

non-repeating...

and infinitely...

available...

to us...
in order for us...
to be able to...
uh...
construct that true belief...
so...
uh...
I'm not...
quite...
sure about it...
but...

looks...
very similar...
to infinitism's tense...
yeah...
interesting...
um...
introduction...
I wasn't...
familiar with this area...
um...
like...
approximate Bayesian computation...
bounded rationality...
the composability of Bayes graphs...
these are...
we get to...
eat our slice of cake...
and recognize...
that there's the rest of the cake...
and the table...
and the world outside of the restaurant...
and so on...
like...
we can just say...

the GM that we made...

was the thermometer...
and a temperature value...
and someone can say...
but what about humidity...
and it's like...
there's a composability...
but...
um...
we're not within the...
positivist...
or the falsificationist...
like...
well...
my model of temperature...
and thermometer...
is positive evidence...
that that's all that's happening...
or...
it's the best model we have...
we're just waiting for it to be disproven...
we can have like...
a Bayesian...
portfolio...
of models...
and all of them...
can be understood as...
maps...
as composable maps...
could this lead...
to...
Bayes Optimal...
experimental design...
of research programs...
like...
statistical power...
the famous...
statistics quotation is like...
something about an autopsy...

like...
the statistician is not gonna diagnose it...
it's...
they come in...
and they tell you what went wrong...

but that's also...
that's not fundamental...
to statistics...
that's actually just a little bit...
of like a joke...
about the practice of research...
which is like...
the biostatistician...
just...
speaking from my own research experience...
they're called in...
for assistance...
on a complex...
already collected data set...
and so it's like...
well...
we had three mice...
in March...
and then we had six mice...
in April...
how do we balance this...
versus the upfront...
conversation...
to help design...
a high statistical power experiment...
and so like...
statistics upfront...
it's gonna...
increase the reproducibility...
and alignability of experiments...
it...
is a sanity check...

if not a formal verification...

that the experiment...

is gonna tell you something...

and...

it...

it allows for...

replicating experiments...

selecting experiments...

based upon like...

their statistical power...

Okay...

following sections...

unpack...

an example of generic inference...

that may be used...

for metabasean inference...

the variational...

Laplace...

with hierarchical models...

then there's gonna be...

a simple recipe...

that's related to the...

one of the other questions...

okay...

!

variational...

may be used...

used for more...

generic likelihood functions...

than those encountered earlier...

which were defined as...

Gaussian...

so previously...

the family...

of functions...

that we were doing...

variational inference with...
was...
Gaussian's...
now...
we can generalize beyond...
using only...
Gaussian distributions...
by saying...
whatever it is...
we're going to do...
a Laplacian approximation...
so even if it's like...

something very strange...
and as...
mentioned...

cohort two...
discussion...
just an hour ago...
Laplace approximation...
has two parameters...
the...

the...
the...
the...

the...
the...

it's guaranteed to capture...
some of the...
variance of the distribution...
for distributions...
with a central tendency...
Gaussian or otherwise...

it can do well...

for distributions...

that are like...

multimodal...

it does not do well...

because it gets tricked...

to finding the mode...

which may only capture...

like a minority...

of the overall bulk...

and again...

the SPM textbook...

lays out...

side by side...

Laplace...

variational Bayes...

non-parametric...

sampling approaches...

from those...

Laplacian approximations...

we're going to do...

variational inference...

so...

the real distribution...

might be...

unfactorizable...

or...

we don't know...

what family of...

it even is...

but...

we know that...

the Laplace approximations...

that we make...

on those distributions...

for sure...

will be amenable...

to variational inference...
and because of the...
quadratic nature...
we can do...
gradient ascent...
so...
just like the ball...
going to the bottom of the bowl...
this is just an inverted...

parabola...
so it's just gradient...

on...
the mountain...
alright...
parametric...

base...
this...
usage...
of...
a matrix...
to represent...
the experimental...
layout...
is heavily...
used...
in SPM...
this is looking...
a lot like...
like a regression...
this is actually...
a little bit of a short...
section 9.4...
this part here...
is...
a little bit of a short...

section 9.4...
section...
this is actually...
a little bit of a short...
section 9.4...
this part here...
comparing the evidence...
for a model...
where the second...
element is allowed to deviate...
from zero...
or the precise belief...
it's zero...
that is actually...
very much...
again...
like regression...
testing...
like...
if you're gonna...
if you have...
if you're testing...
for...
for the effect...
of a given...
factor...
on...
height...
that is related to...
the p-value...
you get...
for height...
is related to...
the model that includes height...
and the model that...
doesn't include height...
and then...
their...

sum of squares...
is compared...
within the appropriate...
statistical test family...
t-test...

f-score...
all giving you...
a p-value...
so...

is very much like...
statistics...
on...
regressions...
one thing...
maybe there...
there's a deeper reading...
where this is...
clearer...
but...
one way to talk about...
empirical bays...
is that like...
well...
first off...
it is...
it's parametric...
you're dealing with parameters...

sometimes that's not too helpful...
but...
the empirical part...
means like...
the way we're gonna get...
our D matrix...
our prior...
is from...

the data...

so like...

let's just say that we have...

we're measuring height...

in the classroom...

we might be able to use...

chapter 6...

to specify...

and furnish...

the model...

but then...

as soon as we make that...

and then we might set...

our prior...

we'll say...

well...

let's have a super loose prior...

let's say that...

uniform...

across all...

you know...

but of course...

who knows...

what the maximum height is...

maybe it's not a human classroom...

so we can't just specify...

a uniform distribution...

across...

you know...

every finite value...

and so...

a way to kind of...

break that...

a priori...

challenge...

where like...

your...

prior...

to be loose enough...

to accommodate...

the full range...

of the possible data...

but also...

sharp enough...

so that you're not just like...

starting like...

on like...

it's just an absurdly...

uninformative prior...

all priors are informative...

even a uniform prior...

is still informative...

the thing that matters...

is to strengthen...

the weakness of the prior...

and the family of the prior...

so what you could do instead...

would be...

you could measure...

10 heights...

or even one...

and then...

use the mean...

and the variance...

of that sample...

empirically...

to set your prior...

so set just the mean...

of the prior...

on height...

as the mean...

that you got from 10...

and then you could just...

over disperse the variance...

so...

if there's somebody...

who's outside of that bound...

now it's like...

you know...

you can be more confident...

so this is widely used...

to take an empirical data set...

and then use...

the data set's...

empirical values...

and summary statistics...

to parameterize...

the generative model...

and then...

the ball goes from there...

and also...

the expectation...

maximization algorithm...

where you have...

a given data set...

and you update the parameters...

in the generative model...

and then you generate data...

compatible with that model...

and continue...

that is...

very close to PEB...

okay...

9.5...

instructions for model-based analysis...

so let's copy these out...

because the question was like...

how is this...

oh...

figure 9.2...

I guess...

section 9.5...

and it's summarized...

in 9.2...

here's the six steps...

and it's summarized...

in 9.2...

here's the six steps...

wait...

this is a little bit of a...

review of the textbook. But we're starting with data collection. Maybe the data already have been
collected. So we're putting aside the situation where we're doing the meta-modeling on ourselves
and then doing experimental design and then collecting the data. We're just going to take
it from the data are collected. A POMDP, again discrete time in this case, is structurally
prepared according to the recipe in chapter 6. POMDPs, would we say, are equivalent to specifying a
likelihood function? Or they embody or they entail a likelihood function? What is the arrow between 2 and 3?
Does merely constructing a POMDP in this format uniquely identify a likelihood function?

Or what work has to happen right here?

Just a thought. We can learn.

Ali, go ahead.

I don't think these arrows necessarily mean that this particular component of the
modeling process necessarily translates perfectly to the other stage. I think they're more likely to be
to show the dependencies among each steps. So in the case of the third stage or the likelihood function,
the construction of the likelihood function, obviously it depends on what our POMDP model
functions. So I'm not sure if we can. I mean, if it were, I mean, if they were 100%
isomorphic, I guess they would be redundant, right? So yeah, I think those arrows show dependencies rather than
translating into each other.

Thanks. I think I broadly agree.

I think that's a good idea.

I think that's a good idea.

I think that's a good idea.

And there might also be some other ways to draw it, like the prior beliefs.

Is this theta?
I think this is using the figure 9-1 ontology.
This theta is big theta.
This is the experimenter's parameters.
The model
is the cognitive model of the entity.
Italic U tilde is the observed data.
So here's the observed data getting passed.
The POMDP and associated likelihood function.
We can't, the likelihood, okay.
The POMDP describes variables in their relationships.
The likelihood function allows us to create P, which is the distribution of observed behavior,
U tilde, conditioned upon experimental parameters data.
Observations, which are in this case, the ones that are provided the stimuli, experimental stimuli,
and the model.
Then the parametric empirical base comes in when we actually get the data.
It comes in in six, but we're getting there.
We're on the path when we have the data flowing from the experiment.
We can then do the model inversion.
Rather than describing the distribution of behavior that would be generated by theta O M,
we're going to invert it so we can talk about the distribution of parameters,
experimental parameters, conditioned upon MOU.
Bayesian equation allows us to have a proportionality between what we really want to know,
this top line, and a, I don't know if I can call it a joint distribution,
but these two multiplied distributions, which is the probability of the experimental parameters given
the model and separating out the probability of U, conditioned on all the rest.
So this is leveraging the sparsity of the inverted model to facilitate a form
that is amenable to linear regression type equations.
This is like Y equals MX plus B.
It's not, but it is.
This is like the same structure.
Y, M, X plus B.
But then these bars and their associated variances, which come from Laplace,
this is the p-value and the effect size.
So for this third modality, the effect size is zero and the variance is such and such.
And then for this one, we could say the p-value or the Bayes factor for this factor mattering,
it's high.
The evidence for this mattering is extremely high.

So this is like, if this were a bar chart and we're, if we were in frequentist statistics land,
and it were like 10 plus or minus one, then we have a Z-score of 10.
We're 10 standard deviations away from zero.
So the p-value of the effect size being greater than zero is
whatever it is for a p-value of a Z-score of 10.
Very low.
So again, this is like where the real
data collection
is going to be happening and interfacing.
So now here's their description.
Collect behavioral data.
Formulate the POMDP.
It takes parameters as input outputs a fully specified but not yet solved POMDP.
Specify likelihood function.
Specify prior beliefs.
Often these will be centered on zero with precisions reflecting plausible ranges.
That's interesting note.
Solve for posterior and model evidence.
Newton is probably a reference there to some gradient,
derivative-based gradient method.
I'm not exactly sure.
Group level analysis.
Treating the estimated parameters for each individual as if they were generated by a second level model.
This is exactly structurally ANOVA.
The group level analysis of variance.
Like, we had people throw the ball.
There was type one and type two a person.
And then the weather type two.
Is there an effect of type?
You make two models.
One of them is a model without type.
One of them is a model with two categories of type.
Then those two models can be compared in terms, in the frequentist world,
in terms of their hierarchical likelihood.
In the Bayesian world, however, a limitation of the hierarchical likelihood ratio test
is that the models must be strictly structurally nested.
They have to reflect direct simplifications or elaborations of each other.
In contrast, Bayesian modeling allows us to use the Bayes factor,

which can compare models that don't need to be strictly nested.
So you could test very structurally different models against each other.
It's kind of like information as a common currency
in the Bayesian approach.
I considered variable one, two, and three.
And then someone else could compare variable four, five, six.
And you can be like, well, how good are those two different models against what we care about?
In terms of informational criterion, AIC, BIC, the Bayes factor.
Whereas if somebody said, well, I did model with one, two, three,
three, and the P value is 0.01.
And I did this with four, five, six, and the P value is 0.01.
Those can't be directly compared because the P value is against like a local null hypothesis.
And there's just like probably other issues.
So that's one huge advantage of the Bayesian.
Like we're kind of, we're past, we're 50 years past the Rubicon or 400 or whatever on Bayes.
But like Bayesian formulations allow for the generative recognition model, the tail of two densities.
So you can specify something that generates and recognizes data.
That's one advantage.
Another advantage is that you have access to all the statistical tools of frequentist statistics.
And you can also access this like informational statistics.
I'm sure there's like way, way, way more rabbit holes there, but like broadly speaking,
instead of taking experimental observations in frequentism and then mapping them onto a T distribution
only, and then saying, well, now because of the P value, let me publish this paper.
In the Bayes world, there are like richer comparisons and like more nuanced penalization
functions of how many parameters versus how much variance explained.
You know, just for example, if anyone thought Bayesians were too confident or if it wasn't the
leading modern strategy for dealing with uncertainty.
So let's see.
This design matrix for a linear model.
The SPM design matrices are like very interesting looking.
So the book, the textbook has many, many, many of these matrices.
They're encoded on a gray scale.
So like what's happening here, here are images through time.
Each row is an image.
Could be an image in a time series.
Could be like different static images.
Here, third column is time.
So this is time point 0, 1, 2, 3, 4, 5, 6.

And then here's condition 1 and 2.

First condition, whether one is white and black, it doesn't matter.

Here, condition, let's just say white is 1.

Condition 1 was in effect.

Condition 2 is not.

And vice versa.

But you could have overlapping conditions.

So it's like a graphical representation.

And like this is also common to see this kind of like waterfall.

Here's patient 1, 2, 3, 4, 5.

Here's the three conditions in a randomized order for the 1, 2, 3, 4, 5 patients.

Here's their global blood flow.

Here's some other measurements, etc.

But it turns out with this, so this summarizes the total design of the experiment.

And it turns out like you just smash it like a matrix multiplication.

Like the design matrix multiplied by the observations.

And one can imagine like if the observations have no relationship with the design matrix,

there's a null hypothesis that you could get from random matrix theory.

If the experimental design strongly influences the observations,

there's some informational outcome.

9.6 in our final few minutes, examples of generative models.

Two experiments outlined in 9.5.

The details are not important.

Oh.

But it's two, um,

eye-secating models.

This is a continuous time.

The second time.

Left.

Tracking a moving target.

Right.

Was a categorical eye tracking.

As we've mentioned and discussed, like,

doing some kind of a webcam,

eye-tracking cognitive model

would be massive.

It's also a very, you know,

sensitive technology, I think.

Because it would be able to determine somebody's cognitive model
and their attention in different ways.
But it would be interesting.
I was going to say,
they actually already do that in,
specifically for, uh,
like, championship, like, first-person shooter games.
To try to catch people, uh, using aim bots.
Like, so they try to predict,
you know, where your eyes are going to be based on the data in the game.
And if you are shooting, you know,
or you're acquiring the target faster than your eyes,
you know, acquire, could have acquired the target then.
Wow.
Yeah.
Well, yeah, interesting.
Maybe even some of the, um, behaviors already
are out there.
So it's not like this is, yeah.
Like, maybe the data already exists.
It's like that moving, moving target one.
Yeah.
Yeah.
And like,
interpreting this as like an occupancy density,
but then having a cognitive model
where that is like an uncertainty reduction or a salience
or relevance observation,
which is implicitly what these usages are.
Disney trailers.
Um,
models of false inference.
Julius Caesar.
We had the Obama.
Now we have the Caesar.
Um, discussion of Bayes optimality and disorders.
Different pathological and neurodiverse conditions.
Addiction, impulsivity, compulsivity,

delusions, hallucinations,

interpersonal personality disorders,

ocular motor syndromes.

Pharmacotherapy, prefrontal syndromes, visual neglect,

disorders of interoceptive inference.

So,

if there's a parameters set that works well,

there's going to be various parameter sets that exhibit different outcomes.

And then as per Foucault, society is the definer of madness and all of that, and the DSM.

And so, you know, asterisk, asterisk, asterisk.

But broadly speaking, these are models of pathology.

Hmm.

Um, we outlined an approach that uses theoretical models described in previous chapters to pose

questions to empirical data.

So, this lets us use active inference as a non-invasive tool to probe the computational processes that individuals use to make decisions.

So funny.

Well, the model is not invasive.

Okay.

I don't think any model is invasive.

Ultimately, the six steps in figure 9.1

is not.

But it's not.

That's an error.

It's figure 9.2.

Six steps in figure 9.2 provide a generic method for designing experiments

to non-invasively interrogate implicit generative models, people,

or other systems used to drive behavior.

That's an opportunity to answer questions about the function of the nervous system in health and indices.

All right.

Thank you, fellows.

Looking forward to the conversation next week as well.

Thank you.

Peace.

Bye.

Thank you.